

Spatiotemporal Signal Propagation in Live-Cell Imaging Data

A Mathematical Formalization

Academic Handout — Systems Biology, University of Warsaw

Companion to `spatiotemporal_signal_propagation.py` and `compare_spatiotemporal_behavior.py`

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1 Biological Setting and Motivation

The data originate from live-cell fluorescence microscopy experiments on **MCF10A** cells (a non-tumorigenic human mammary epithelial line). Each cell carries two fluorescent kinase-activity reporters:

- **ErkKTR** — an ERK kinase translocation reporter whose cytoplasmic-to-nuclear fluorescence ratio (`ERKKTR_ratio`) is a real-time proxy for ERK activity.
- **FoxO3A** — a nuclear/cytoplasmic ratio reporter for the PI3K→AKT→FoxO3A signalling axis.

Cells were growth-factor-deprived for 48 h prior to acquisition, seeded in 96-well plates ($\sim 25\,000$ cells/well), and imaged every $\Delta t = 5$ min for a total of $T_{\max} = 24$ h (up to 288 frames). The dataset spans six replicate experiments and 28 imaging sites assigned to different *PI3K-pathway genetic backgrounds*: wild-type (WT), **AKT1** E17K, **PIK3CA** E545K, **PIK3CA** H1047R, and **PTEN** deletion.

Core biological question. *When a cell shows a sudden, large increase (“jump”) in ERK or FoxO3A activity, are its spatial neighbours more likely to show their own large activity increase shortly afterwards?* If yes, this suggests local cell-to-cell signalling or coordinated behaviour, and we want to quantify how strongly this effect depends on the genetic background.

2 Data Model: Cells, Tracks, and Nodes

2.1 Primary observables

Let $\mathcal{C}$ be the set of all *cell tracks* in one experiment-site block. A track $k \in \mathcal{C}$ is a sequence of time-resolved observations of the *same physical cell*. Let $T_k \subseteq \{1, 2, \dots, T_{\max}\}$ be the set of frames during which track k is observed.

Definition 2.1 (Node). A *node* is a pair $v = (k, t)$ with $k \in \mathcal{C}$ and $t \in T_k$. The set of all nodes is

$$V = \{(k, t) \mid k \in \mathcal{C}, t \in T_k\}.$$

Each node (k, t) carries:

$\mathbf{x}(k, t) = (x_{k,t}, y_{k,t}) \in \mathbb{R}^2$	nuclear centroid (image pixels),
$S_k(t) \in \mathbb{R}_{>0}$	fluorescence ratio (signal channel),
$A(k, t) \in \mathbb{R}_{>0}$	nuclear area (pixels).

Remark 2.1. Physical time in hours is $t_h = t \cdot \Delta t / 60$ with $\Delta t = 5$ min.

3 Graph Construction

The scripts build a *spatiotemporal graph* $G = (V, E_{\text{sp}} \cup E_{\text{tm}})$ with two disjoint families of edges.

3.1 Spatial edges

Definition 3.1 (Spatial neighbourhood). For node (k, t) and radius $r > 0$, the *spatial neighbourhood* is

$$\mathcal{N}_k(t) = \{(k', t) \mid k' \neq k, \|\mathbf{x}(k, t) - \mathbf{x}(k', t)\|_2 \leq r\}.$$

A *spatial edge* exists between (k, t) and (k', t) whenever $(k', t) \in \mathcal{N}_k(t)$, i.e. both cells are within distance r of each other *at the same frame* t . The full set is

$$E_{\text{sp}} = \{\{(k, t), (k', t)\} \mid (k', t) \in \mathcal{N}_k(t)\}.$$

The default radius is $r = 60$ image-coordinate units. Spatial edges are *undirected* and *frame-local*: a cell pair is connected only at the frame(s) at which they are close.

Remark 3.1 (Efficient computation via k -d trees). At each frame t , all nuclear centroids are inserted into a k -d tree (a balanced binary space-partition structure). Querying all pairs within radius r costs $\mathcal{O}(|V_t| \log |V_t| + |E_{\text{sp}, t}|)$ per frame, where $V_t = \{(k, t) \mid t \in T_k\}$. The k -d tree approach is therefore far more efficient than naively checking all $\binom{|V_t|}{2}$ pairs.

3.2 Temporal edges

Definition 3.2 (Temporal edge). For each track k and each pair of *consecutive observed frames* $t, t' \in T_k$ with $t < t'$ and no intervening observation in T_k , there is a *directed temporal edge* $(k, t) \rightarrow (k, t')$. The set is

$$E_{\text{tm}} = \{(k, t) \rightarrow (k, t') \mid k \in \mathcal{C}, t' = \min_{s \in T_k, s > t} s\}.$$

Each temporal edge records the frame gap $\delta = t' - t$ (nominally $\delta = 1$, but tracking gaps can exceed one frame) and the physical time gap $\delta \cdot \Delta t$ in minutes.

Remark 3.2 (Graph structure). The temporal edges alone form a collection of directed paths — one per track — corresponding to the trajectories of individual cells over time. Adding spatial edges introduces *cross-track* connections at each frame, converting the path collection into a richer spatiotemporal graph.

4 Signal Dynamics and Jump Detection

4.1 First difference (signal delta)

Along each temporal edge $(k, t) \rightarrow (k, t')$ in E_{tm} , define the *signal increment*

$$\Delta S_k(t') = S_k(t') - S_k(t).$$

When (k, t) is the first observed frame of track k (i.e. (k, t) has no incoming temporal edge), $\Delta S_k(t)$ is undefined (NaN).

4.2 Data-driven jump threshold

Definition 4.1 (Jump threshold). Let

$$\mathcal{D}^+ = \{\Delta S_k(t) \mid (k, t) \in V, \Delta S_k(t) > 0\}$$

be the multiset of all *positive* signal increments in the block. For a chosen quantile $q \in (0, 1)$, the *data-driven jump threshold* is

$$\theta_q = Q_q(\mathcal{D}^+),$$

where Q_q is the q -th empirical quantile. The default is $q = 0.90$, so $\theta_{0.90}$ equals the 90th percentile of all upward steps in the block. A user-supplied fixed threshold overrides this per-block estimate.

4.3 Jump event indicator

Definition 4.2 (Jump event). For node (k, t) , the *jump event indicator* is

$$\mathcal{J}_k(t) = \mathbf{1}[\Delta S_k(t) \geq \theta],$$

using the convention $\mathcal{J}_k(t) = 0$ when $\Delta S_k(t)$ is undefined.

Intuitively, $\mathcal{J}_k(t) = 1$ means cell k at frame t shows a “sharp upward step” in signal activity that is in the top $(1 - q)$ fraction of all positive changes observed across the block.

5 Spatiotemporal Exposure and Future Response

5.1 Future self-jump indicator

Definition 5.1 (Future self-jump). For node (k, t) and a *future window* $W \in \mathbb{Z}_{>0}$ (frames), the *future self-jump indicator* is

$$\mathcal{F}_k(t) = \mathbf{1}[\exists t' \in T_k : t < t' \leq t + W \text{ and } \mathcal{J}_k(t') = 1].$$

Thus $\mathcal{F}_k(t) = 1$ if and only if cell k itself undergoes at least one jump within the next W frames. The default is $W = 3$, corresponding to a 15-minute look-ahead window.

The future self-jump is computed by a forward scan along each track: for every node (k, t) , the algorithm inspects subsequent frames $t + 1, t + 2, \dots$ until either a jump is found or the window W is exhausted.

5.2 Spatial exposure

Definition 5.2 (Neighbourhood statistics and exposure). For node (k, t) , define:

$$n(k, t) = |\mathcal{N}_k(t)| \quad (\text{spatial neighbour count}), \quad (1)$$

$$m(k, t) = \sum_{(k', t) \in \mathcal{N}_k(t)} \mathcal{J}_{k'}(t) \quad (\text{number of jumping neighbours at frame } t), \quad (2)$$

$$\bar{S}(k, t) = \frac{1}{n(k, t)} \sum_{(k', t) \in \mathcal{N}_k(t)} S_{k'}(t) \quad (\text{mean neighbour signal; undefined if } n(k, t) = 0). \quad (3)$$

The *exposure indicator* is

$$E_k(t) = \mathbf{1}[m(k, t) > 0],$$

equal to 1 when at least one spatial neighbour of cell k is currently undergoing a jump.

Remark 5.1. Because spatial edges are rebuilt per frame, $\mathcal{N}_k(t)$ and hence $n(k, t)$ can differ at each time point as cells migrate within the imaging field.

6 Propagation Statistics

The central question is whether exposure to a jumping neighbour *right now* is predictive of a future self-jump. We operationalise this with two classical epidemiological estimators applied to cell-time nodes.

6.1 Exposed and unexposed populations

Partition V into two disjoint subsets:

$$V_{\text{exp}} = \{(k, t) \in V \mid E_k(t) = 1\}, \quad V_{\text{unexp}} = \{(k, t) \in V \mid E_k(t) = 0\}.$$

6.2 Conditional future-jump rates

Define the *conditional future-jump rates*

$$p_{\text{exp}} = \Pr(\mathcal{F}_k(t) = 1 \mid E_k(t) = 1) \approx \frac{1}{|V_{\text{exp}}|} \sum_{(k,t) \in V_{\text{exp}}} \mathcal{F}_k(t), \quad (4)$$

$$p_{\text{unexp}} = \Pr(\mathcal{F}_k(t) = 1 \mid E_k(t) = 0) \approx \frac{1}{|V_{\text{unexp}}|} \sum_{(k,t) \in V_{\text{unexp}}} \mathcal{F}_k(t). \quad (5)$$

6.3 Risk difference and relative risk

Definition 6.1 (Risk difference and relative risk).

$$\text{RD} = p_{\text{exp}} - p_{\text{unexp}}, \quad (6)$$

$$\text{RR} = \frac{p_{\text{exp}}}{p_{\text{unexp}}} \quad (p_{\text{unexp}} > 0). \quad (7)$$

Interpretation.

- RD = 0 (or RR = 1): neighbour jumping has no association with future self-jumping.
- RD > 0 (or RR > 1): exposed cell-time points are more likely to be followed by a self-jump. An RD of 0.05 is a 5 percentage-point absolute increase. An RR of 1.76 means exposed cells are $\approx 1.76\times$ more likely to jump in the near future.
- RD < 0 (or RR < 1): exposure is associated with a *lower* future-jump probability (potential suppression).

6.4 Empirical example

For the WT condition in this dataset (one block, ERK signal):

Mutation	$ V $	$ E_{\text{sp}} $	θ	p_{exp}	p_{unexp}	RR
WT	356,632	2,389,504	0.034	0.131	0.075	1.756

Wild-type cells whose neighbour jumps *now* are roughly 75% more likely to show their own ERK jump within the next 15 minutes.

Remark 6.1 (Statistical note). Equations (4) and (5) treat every node as an independent observation. This is a simplification: nodes from the same track are serially correlated, and spatially adjacent nodes share neighbourhood structure. A rigorous inference (e.g. permutation test with tracks as units, or a mixed-effects log-binomial regression) is needed to assign valid p -values to RR.

7 Multi-Block Comparison Framework

7.1 Block-level summaries

An *experiment-site block* $\mathcal{B}_{e,s}$ is the subset of all observations belonging to experiment e and imaging site s . Let $\mathcal{B} = \{\mathcal{B}_{e_1,s_1}, \dots, \mathcal{B}_{e_B,s_B}\}$ be the collection of blocks remaining after filtering (e.g. removing the ErbB2 condition from the comparison).

Running the single-block algorithm of Section 5 on each block b yields the summary tuple

$$\phi(b) = (n_b, |E_{\text{sp},b}|, |E_{\text{tm},b}|, \theta_b, p_{\text{exp}}^{(b)}, p_{\text{unexp}}^{(b)}, \text{RD}^{(b)}, \text{RR}^{(b)}),$$

where $n_b = |V_b|$ and θ_b is computed per-block when no global threshold is supplied.

7.2 Group-level aggregation

Let $\gamma : \mathcal{B} \rightarrow \Gamma$ assign each block to a comparison group $g \in \Gamma$ (default: $\Gamma = \text{set of distinct mutation labels}$). For each group g , let $\mathcal{B}_g = \{b \in \mathcal{B} \mid \gamma(b) = g\}$ and define

$$\overline{\text{RR}}(g) = \frac{1}{|\mathcal{B}_g|} \sum_{b \in \mathcal{B}_g} \text{RR}^{(b)}, \quad (8)$$

$$\widetilde{\text{RR}}(g) = \text{median}_{b \in \mathcal{B}_g} \{\text{RR}^{(b)}\}, \quad (9)$$

$$\overline{\text{RD}}(g) = \frac{1}{|\mathcal{B}_g|} \sum_{b \in \mathcal{B}_g} \text{RD}^{(b)}. \quad (10)$$

Groups are sorted by $\overline{\text{RR}}(g)$ in descending order. The full aggregated table is saved to `group_level_summary.csv`.

Remark 7.1 (Automatic grouping). When the metadata column **Conditions** takes only a single value across all sites (as in this dataset, where all wells share the same growth-factor deprived condition), the script automatically uses **Mutation** as the comparison grouping.

8 Formal Summary: The Algorithm

Algorithm 1. Single-block spatiotemporal propagation analysis.

Input: block $\mathcal{B}_{e,s}$; spatial radius r ; quantile q ; future window W ; optional fixed threshold θ .

Output: summary tuple $\phi(\mathcal{B}_{e,s})$.

1. (*Signal delta*) For each $(k, t) \in V$: compute $\Delta S_k(t) = S_k(t) - S_k(t_{\text{prev}})$ where $t_{\text{prev}} = \max\{s \in T_k : s < t\}$; set to NaN if no predecessor exists.
2. (*Threshold*) If no fixed θ : set $\theta \leftarrow Q_q(\{\Delta S_k(t) > 0\})$.
3. (*Jump events*) $\mathcal{J}_k(t) \leftarrow \mathbf{1}[\Delta S_k(t) \geq \theta]$ for all $(k, t) \in V$.
4. (*Spatial edges*) For each frame t , build a k -d tree on $\{\mathbf{x}(k, t)\}_{t \in T_k}$ and enumerate all pairs within radius r to form $E_{\text{sp}, t}$; set $E_{\text{sp}} = \bigcup_t E_{\text{sp}, t}$.
5. (*Temporal edges*) For each track k , connect consecutive observations to form E_{tm} .
6. (*Future self-jump*) For each $(k, t) \in V$: scan frames $t + 1, t + 2, \dots, t + W$ in T_k and set $\mathcal{F}_k(t) \leftarrow 1$ if any of them is a jump event.
7. (*Exposure*) For each $(k, t) \in V$: compute $m(k, t)$ via E_{sp} ; set $E_k(t) \leftarrow \mathbf{1}[m(k, t) > 0]$.
8. (*Summary*) Compute $p_{\text{exp}}, p_{\text{unexp}}$ via Eqs. (4)–(5); return $\text{RD} = p_{\text{exp}} - p_{\text{unexp}}$ and $\text{RR} = p_{\text{exp}}/p_{\text{unexp}}$.

9 Discussion and Connections

9.1 What this analysis captures

The analysis measures the *empirical association* between a current neighbourhood jump and a future self-jump, summarised by RR and RD. An $\text{RR} > 1$ is consistent with—but does not prove—mechanisms such as:

- direct paracrine signalling (e.g. secreted growth factors) between physically adjacent cells;

- a common upstream spatial stimulus (e.g. a localised wave of extracellular signal) that activates nearby cells in sequence;
- spatially correlated noise in the shared microenvironment.

9.2 Key assumptions and caveats

- (a) **Node independence.** Treating each (k, t) as i.i.d. inflates the effective sample size and invalidates standard confidence intervals. A valid uncertainty estimate uses the *track* as the primary sampling unit.
- (b) **Threshold sensitivity.** The binary jump indicator depends strongly on θ . The per-block quantile threshold makes blocks non-comparable unless a global θ is fixed.
- (c) **Temporal resolution.** The exposure variable $E_k(t)$ captures *concurrent* neighbour jumps only. A lagged-exposure analysis (e.g. $E_k(t - 1)$ or $E_k(t - 2)$) would separate temporal precedence from simultaneity.
- (d) **Spatial confounding.** Spatially adjacent cells may share microenvironmental conditions (local ECM density, medium gradients) that independently affect signalling, independent of direct cell–cell communication.

9.3 Connections to other mathematical frameworks

Temporal/dynamic networks. $G = (V, E_{\text{sp}} \cup E_{\text{tm}})$ is a *dynamic graph* in which spatial edge sets change at each time step while temporal edges encode causal ordering within tracks.

Geometric random graphs. The spatial edge set $E_{\text{sp},t}$ at any frame t is precisely the edge set of the *unit-disc graph* on centres $\{\mathbf{x}(k, t)\}$, i.e. the intersection graph of discs of radius $r/2$.

Log-binomial regression. The RR estimator can be extended to a fully adjusted model:

$$\log \Pr(\mathcal{F}_k(t) = 1) = \alpha + \beta E_k(t) + \boldsymbol{\gamma}^\top \mathbf{z}(k, t),$$

where $\mathbf{z}(k, t)$ contains covariates such as nuclear size, baseline signal, or frame number. Then $e^{\hat{\beta}} \approx \text{RR}$ conditionally on those covariates.

Point process perspective. The sequence of jump events along a track $\{t \in T_k : \mathcal{J}_k(t) = 1\}$ is a discrete *point process* on the time axis; the analysis asks whether the rate of this process is elevated following a jump in a spatially proximal process.

Appendix: Notation Summary

Symbol	Meaning
V	Set of all (cell, frame) nodes
E_{sp}	Spatial edges (same frame, within radius r)
E_{tm}	Temporal edges (same track, consecutive frames)
k	Track (cell) index
t	Frame index (integer, $1 \leq t \leq T_{\text{max}}$)
T_k	Set of frames at which track k is observed
$\mathbf{x}(k, t)$	Nuclear centroid (x, y) in image pixels
$S_k(t)$	Signal ratio (ERKKTR or FoxO3A) at node (k, t)
$\Delta S_k(t)$	Signal increment: $S_k(t) - S_k(t_{\text{prev}})$
θ	Jump threshold (fixed or data-driven)
q	Quantile for data-driven θ (default: 0.90)
$\mathcal{J}_k(t)$	Jump event: $\mathbf{1}[\Delta S_k(t) \geq \theta]$
r	Spatial radius for neighbourhood (default: 60 units)
$\mathcal{N}_k(t)$	Spatial neighbourhood of node (k, t)
$n(k, t)$	Neighbour count: $ \mathcal{N}_k(t) $
$m(k, t)$	Jumping-neighbour count at frame t
$\bar{S}(k, t)$	Mean signal of spatial neighbours
$E_k(t)$	Exposure indicator: $\mathbf{1}[m(k, t) > 0]$
W	Future window in frames (default: 3, i.e. 15 min)
$\mathcal{F}_k(t)$	Future self-jump: $\mathbf{1}[\exists t' \in (t, t+W] : \mathcal{J}_k(t') = 1]$
$p_{\text{exp}}, p_{\text{unexp}}$	Conditional future-jump rates (exposed / unexposed)
RD	Risk difference: $p_{\text{exp}} - p_{\text{unexp}}$
RR	Relative risk: $p_{\text{exp}}/p_{\text{unexp}}$
$\mathcal{B}_{e,s}$	Experiment-site block (experiment e , site s)
$\overline{\text{RR}}(g)$	Mean relative risk aggregated over group g